ReflectGAN: Modeling Vegetation Effects for Soil Carbon Estimation From Satellite Imagery

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***Abstract*—Soil organic carbon (SOC) is a critical indicator of soil health, but its accurate estimation from satellite imagery is hindered in vegetated regions due to spectral contamination from plant cover, which obscures soil reflectance and reduces model reliability. This study proposes the reflectance transformation generative adversarial network (ReflectGAN), a novel paired GAN-based framework designed to reconstruct accurate bare-soil reflectance from vegetated soil satellite observations. By learning the spectral transformation between vegetated and bare-soil reflectance, ReflectGAN facilitates more precise SOC estimation under mixed land cover conditions. Using the land use/land cover area frame survey (LUCAS) 2018 dataset and corresponding Landsat 8 imagery, we trained multiple learning-based models on both original and ReflectGAN-reconstructed reflectance inputs. Models trained on ReflectGAN outputs consistently outperformed those using existing vegetation correction methods. For example, the best-performing model (RF) achieved an R^2 of 0.54, root-mean-squared error (RMSE) of 3.95, and residual prediction deviation (RPD) of 2.07 when applied to the ReflectGAN-generated signals, representing a 35% increase in R^2, a 43% reduction in RMSE, and a 43% improvement in RPD compared to the best existing method (PMM-SU). The performance of the models with ReflectGAN is also better compared to their counterparts when applied to another dataset, i.e., Sentinel-2 imagery. These findings demonstrate the potential of ReflectGAN to improve SOC estimation accuracy in vegetated landscapes, supporting more reliable soil monitoring.**

***Index Terms*—Generative adversarial network (GAN), Landsat 8, remote sensing, Sentinel-2, soil organic carbon (SOC).**

I. Introduction

SOIL organic carbon (SOC) is a fundamental indicator of soil health, influencing agricultural productivity, carbon sequestration, improved soil moisture retention, and overall ecosystem sustainability. Accurate estimation of SOC is essential for promoting sustainable agriculture, improving soil management practices, and monitoring environmental changes [1], [2]. Traditional methods for estimating SOC rely on laboratory-based soil analyses, which, although precise, are labor-intensive, costly, and limited in spatial coverage [3], [4]. These constraints have led to growing adoption of remote sensing technologies as a scalable and cost-effective alternative to conventional field sampling and laboratory analysis.

Laboratory-based hyperspectral imaging (HSI) provides a powerful tool for SOC estimation by offering high spatial and spectral resolution, enabling detailed analysis of soil properties without the need for destructive sampling [5], [6], [7]. Numerous studies have validated the effectiveness of HSI in accurately estimating SOC levels [7], [8]. However, the widespread deployment of HSI is constrained by the high cost of equipment and limited accessibility, making it impractical for large-scale applications. These limitations have prompted interest in more affordable alternatives, such as visual-band multispectral imaging, which can achieve competitive SOC estimation accuracy at substantially lower costs [9], [10]. Still, both HSI and multispectral imaging systems are typically confined to localized areas. To address these spatial constraints, satellite-based remote sensing has emerged as a promising approach for large-scale SOC estimation, offering broader spatial coverage and cost-effective monitoring across diverse landscapes.

Multispectral satellite platforms such as Sentinel-2 (S2) and Landsat 8 (L8) offer broad spatial coverage and public accessibility, making them attractive for large-scale SOC estimation. However, their estimation accuracy often lags behind hyperspectral approaches due to environmental noise, limited spectral resolution, and the need for extensive preprocessing [11], [12], [13], [14]. To mitigate these challenges, recent studies have integrated supplementary features—such as vegetation indices (VIs), soil indices (SIs), and tasseled cap transformation (TCT) components—alongside raw spectral bands. This feature augmentation has been shown to improve the performance of learning-based models by enhancing their ability to capture the complex variability associated with SOC [13].

Nevertheless, most existing studies have primarily focused on bare-soil conditions, where minimal or no vegetation is present. This emphasis stems from the fact that vegetation substantially alters surface reflectance, masking the underlying

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soil characteristics and complicating SOC estimation. Consequently, models are often trained exclusively on bare-soil samples to minimize spectral interference. However, in real-world agricultural and natural landscapes, bare-soil exposures are seasonal and limited, underscoring the importance of developing methods that can reliably estimate SOC under vegetated conditions—an area that remains insufficiently explored.

To address vegetation interference in SOC estimation, existing vegetation correction strategies can be broadly categorized into three paradigms: 1) correction-based approaches that attempt to suppress vegetation effects [15]; 2) elimination-based approaches that discard vegetation-affected samples [16]; and 3) reconstruction-based approaches that explicitly recover bare-soil-equivalent reflectance [17], [18]. We review these paradigms below and highlight their limitations for large-scale SOC estimation under vegetation.

Traditional approaches to vegetation correction in satellite-based SOC estimation include vegetation index-based techniques, spectral unmixing models, and radiative transfer frameworks. VIs, such as NDVI and SAVI, are commonly used to estimate vegetation cover and partially reduce its impact on reflectance measurements [19]. While computationally efficient, these methods do not recover the true underlying soil reflectance and often yield poor SOC estimation performance in densely vegetated areas [20]. More physically grounded models, such as PROSAIL [21], simulate canopy–soil reflectance interactions using radiative transfer theory. However, their application is limited by the need for numerous biophysical parameters (e.g., leaf area index, leaf angle, soil moisture), which are difficult to obtain at scale.

Spectral unmixing approaches, such as spectral mixture analysis (SMA) [22], [23] and adaptive SMA (ASMA) [24], attempt to separate soil and vegetation components by estimating fractional abundances of endmembers. While conceptually appealing, these methods assume linear mixing models and often rely on idealized endmembers that may not adequately capture the nonlinear reflectance behavior of real-world landscapes. To address these limitations, several nonlinear unmixing models have been proposed, which are better suited for complex environments where reflectance is inherently nonlinear. These include autoencoder-based networks grounded in the multilinear mixing model, which effectively model complex nonlinear interactions in hyperspectral data [25], model-based deep autoencoder networks that incorporate prior knowledge of the mixing process to enhance unmixing performance [26], and interpretable deep learning (DL) methods that account for nonlinearity and endmember variability through a probabilistic variational framework [27]. Probabilistic extensions like the probabilistic mixture model-based spectral unmixing (PMM-SU) [28], [29] account for uncertainty in unmixing but still treat vegetation as a contaminant to be separated rather than learning to reconstruct a soil-equivalent reflectance.

Although these advanced DL unmixing models demonstrate strong capability in handling nonlinear spectral mixtures, they are primarily designed for abundance estimation or latent representation learning, rather than for reconstructing physically consistent bare-soil reflectance required for downstream SOC estimation [30], [31]. As a result, alternative strategies have been explored.

An alternative to spectral unmixing is to bypass contaminated signals entirely by identifying and excluding vegetation-affected samples. Recent approaches, such as transformer-guided noise detection, aim to filter out anomalous reflectance patterns that deviate from expected soil signals [15]. Although effective in improving training quality, these techniques reduce sample availability and limit the model's applicability in persistently vegetated regions.

Reconstruction-based methods offer a promising solution by learning to generate bare-soil reflectance directly from vegetation-contaminated inputs. Some attention-based DL models have been proposed to model soil–vegetation interactions [17], but they do not explicitly reconstruct bare-soil reflectance. As a result, residual vegetation signals may persist in the output, reducing interpretability and potentially degrading SOC estimation accuracy. Explicit reconstruction is essential to ensure that the reflectance conforms to bare-soil spectral patterns, thereby enhancing the reliability of the SOC model and facilitating comparisons with bare-soil-based studies.

Generative adversarial networks (GANs) [32] offer a data-driven approach for spectral reconstruction. While Pix2Pix [33] and CycleGAN [34] have been successful in visual tasks, their objective functions typically prioritize image realism and cyclic consistency over spectral fidelity, making them less suitable for reflectance recovery in geophysical applications.

Beyond image-to-image translation, GANs have been widely adopted in remote sensing tasks such as super-resolution [35] and synthetic scene generation [36], [37]. However, most of these models prioritize visual or spatial details rather than the accurate reconstruction of spectral profiles. StyleGAN and its variants [38], [39], [40] excel in generating photorealistic images from latent vectors but lack mechanisms for structured spectral translation. More recently, GANs have shown potential in hyperspectral unmixing, with models like HyperGAN [41], Wasserstein GAN-based spectral networks [42], and adversarial learning frameworks for endmember-abundance separation [43]. A 3D-GAN was also introduced for spectral unmixing in planetary datasets, improving robustness and generalization under data scarcity [44]. However, these methods primarily estimate abundance fractions or map latent representations rather than reconstructing full reflectance profiles—limiting their utility for direct SOC estimation.

For large-scale SOC mapping, vegetation correction methods must satisfy three requirements: scalability to large satellite datasets, the ability to model nonlinear soil–vegetation spectral interactions, and preservation of physically meaningful soil reflectance for downstream regression [18], [25]. Existing learning-based approaches are limited in this context because they may discard vegetation-affected samples, estimate latent representations without reconstructing soil-equivalent spectra, or lack explicit constraints to preserve spectral fidelity [30], [45]. These limitations motivate a generative framework that learns a direct, physically consistent mapping from vegetation-contaminated reflectance to bare-soil

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reflectance. Among generative models, GANs are particularly well suited to this task because the adversarial discriminator explicitly enforces realism in the generated reflectance, encouraging the generator to produce spectra that are statistically indistinguishable from real bare-soil reflectance rather than merely minimizing reconstruction error.

To overcome these limitations, this study proposes reflectance transformation GAN (ReflectGAN)—a novel GAN architecture tailored to reconstruct full bare-soil reflectance profiles from vegetation-affected inputs. Unlike traditional unmixing-focused GANs, ReflectGAN does not aim to extract fractional abundances. Instead, it learns a direct mapping between paired vegetated and bare-soil reflectance spectra. By integrating residual learning and spectral conditioning, ReflectGAN ensures that the reconstructed spectra retain physical fidelity, enabling robust SOC estimation even in moderately to densely vegetated regions.

ReflectGAN learns the systematic relationship between vegetated and bare-soil reflectance using paired training examples, explicitly modeling the structured spectral distortion introduced by vegetation. Rather than treating vegetation as random noise, ReflectGAN conditions its generation on the input vegetated reflectance and incorporates residual learning blocks within the generator to recover the underlying bare-soil characteristics. This enables the model to disentangle vegetation interference and reconstruct realistic soil reflectance patterns, even in areas with moderate to dense canopy cover. By generating more accurate bare-soil spectra, ReflectGAN significantly enhances SOC estimation in environments where vegetation has traditionally hindered reliable analysis.

The main contributions of this study are summarized as follows.

1) *Development of a GAN Architecture for Spectral Correction:* ReflectGAN introduces a novel paired GAN-based framework specifically designed to transform vegetation-contaminated satellite reflectance into bare-soil reflectance. The model is tailored for spectral correction, addressing vegetation-induced spectral distortion, thereby improving the accuracy of SOC estimation from satellite data.
2) *Overcoming the Limitations of Conventional and Unmixing-Focused GAN Models:* Existing GAN architectures, such as Pix2Pix and StyleGAN [33], [38], primarily focus on visual realism, often compromising spectral fidelity. Recent GAN-based nonlinear unmixing models mainly aim to estimate abundance fractions or classify spectral mixtures in generative space [41], [42], [44], rather than reconstructing full, physically consistent reflectance profiles. In contrast, ReflectGAN is explicitly designed to recover complete bare-soil reflectance by integrating residual learning to maintain spectral consistency and employing spectral conditioning to ensure that generated reflectance aligns with physically accurate bare-soil spectra, thereby preserving physical integrity.
3) *Enhanced SOC Estimation in Vegetated Landscapes:* Experiments demonstrate that learning-based models trained on ReflectGAN-reconstructed reflectance consistently outperform those using uncorrected inputs, particularly in vegetated regions. Compared to transformer-based noise filtering approaches [15], ReflectGAN supports broader spatial coverage by reconstructing rather than discarding vegetation-affected samples.

Through these contributions, ReflectGAN advances the current state of remote sensing-based soil analysis by offering a practical and scalable solution for SOC estimation under real-world, mixed land cover conditions.

The remainder of this article is organized as follows. Section II describes the dataset preparation, including soil sample selection and satellite data processing. Section III details the proposed ReflectGAN architecture. Section IV outlines the overall SOC estimation method. Section V presents the experimental results, while Section VI provides a discussion on key findings, limitations, and future research. Section VII concludes the article.

II. Dataset Preparation

A. Study Area and Soil Data Collection

This study utilized the land use/land cover area frame survey (LUCAS) 2018 topsoil dataset, a standardized soil monitoring initiative coordinated by Eurostat covering EU member states. The dataset comprises 18 984 georeferenced topsoil samples with comprehensive physical and chemical properties, including pH ($CaCl_2$ and H_2O), electrical conductivity (EC), SOC, $CaCO_3$, phosphorus (P), nitrogen (N), and potassium (K), all sourced from European Soil Data Centre (ESDAC). Each sample is associated with precise geolocation (`LUCAS-SOIL-2018.shp`), enabling accurate alignment with satellite imagery for remote sensing applications.

To support reliable SOC estimation from remote sensing, we selected a subset of samples spanning a gradient of vegetation cover, specifically targeting bare-soil and barley-vegetated areas. Sample selection was guided by their associated normalized difference vegetation index (NDVI) values, constrained within the range $0 < \text{NDVI} < 0.98$, capturing conditions from completely bare soil to dense crop cover. This stratified sampling strategy facilitates the focused evaluation of the proposed ReflectGAN framework, which aims to reconstruct accurate soil reflectance by mitigating vegetation-induced spectral distortions in satellite observations.

Table I summarizes the descriptive statistics of SOC in the selected dataset. SOC values exhibit substantial variability, ranging from 2.5 to 30 g/kg, reflecting the diverse soil and land use conditions across the study area. The coefficient of variation (CV) of 47% underscores the heterogeneity of the dataset—an important factor for assessing the robustness of vegetation correction approaches in SOC estimation pipelines.

B. Landsat 8 Data Collection and Preprocessing

To integrate satellite-derived spectral information with ground-truth SOC data, we utilized imagery from the Landsat 8 operational land imager (OLI) sensor. Images were acquired via the USGS Earth Explorer portal (https://earthexplorer.usgs.gov/) by spatially matching each soil

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TABLE I
DESCRIPTIVE STATISTICAL PARAMETERS FOR THE SOC DATASET INVESTIGATED IN THIS STUDY, ILLUSTRATING THE VARIABILITY AND DISTRIBUTION OF SOC VALUES USED FOR MODEL TRAINING AND EVALUATION

Soil type	NDVI range	S. No.	Min	Max	Mean	Median	Std.	CV(%)
Mixed	$0 < \text{NDVI} < 0.98$	554	2.5	30	14.17	12.85	6.66	47

sample's coordinates. Temporal filtering was applied to ensure consistency with field measurements, restricting image acquisition to within ±15 days of the sampling date. Additionally, only images with less than 10% cloud cover were selected to minimize atmospheric interference and ensure reflectance reliability.

Among the available L8 bands, we focused on Bands 1 through 7 (B1–B7), which have demonstrated relevance in soil and vegetation studies [13], [46]. The selection of L8 bands B1–B7 was motivated by their established sensitivity to soil and vegetation spectral characteristics relevant to SOC estimation. The visible bands (B2–B4) capture soil color and organic matter-related brightness variations, while the near-infrared and shortwave infrared bands (B5–B7) are sensitive to soil moisture, surface roughness, mineral composition, and vegetation structure. These bands jointly provide complementary information on soil background properties and vegetation-induced spectral masking, making them well-suited for learning-based soil reflectance reconstruction and SOC modeling. Thermal bands were excluded as they primarily encode surface temperature rather than soil compositional information and have limited relevance for reflectance-based SOC estimation. Thermal and atmospheric bands (e.g., B10–B11) were excluded to retain only spectrally informative features for SOC estimation [47].

The satellite images underwent a multistep preprocessing workflow. First, radiometric calibration converted digital number (DN) values into top-of-atmosphere reflectance, incorporating solar angle and scene metadata [48]. Atmospheric correction was then applied using the fast line-of-sight atmospheric analysis of spectral hypercubes (FLAASHs) algorithm [49], correcting for scattering and absorption to produce surface reflectance.

To enhance spatial resolution, Gram–Schmidt pansharpening was performed by fusing the 30-m multispectral bands with the 15-m panchromatic band (B8) [50]. This step preserved spectral fidelity while improving the spatial detail necessary for capturing subtle soil variability.

Preprocessing tasks were executed in ENVI (v5.6.1), while georeferencing and reflectance extraction were performed using ArcGIS Pro (v2.8.0). Final surface reflectance values from the selected seven bands were extracted at each sample's precise geolocation and compiled into an Excel spreadsheet for subsequent analysis.

C. Landsat 8 Image Transformation and VIs

The inclusion of vegetation and SIs was motivated by their ability to explicitly encode biophysical properties that govern the interaction between vegetation cover and soil reflectance. VIs such as NDVI, SAVI, MSAVI, and EVI quantify vegetation density and canopy effects that obscure soil spectral signals, while soil-related indices (e.g., BI, DSI, and CI) emphasize surface brightness, dryness, and texture characteristics that are closely linked to SOC variability. Together, these indices provide complementary, physically interpretable features that enhance discrimination between vegetation-dominated and soil-dominated spectral responses, thereby supporting both SOC regression and the learning of vegetation-to-soil reflectance transformations in ReflectGAN.

To enhance the spectral feature space and improve SOC modeling, a series of vegetation and SIs was derived from the reflectance values of the seven selected L8 bands. VIs are widely used to assess canopy density and photosynthetic activity, while SIs help characterize surface attributes such as texture, moisture, and color—factors closely associated with SOC variability.

In this study, we calculated several commonly used VIs, including the ratio vegetation index (RVI), NDVI, green NDVI (GNDVI), enhanced vegetation index (EVI), soil-adjusted vegetation index (SAVI), and modified SAVI (MSAVI). Additionally, a set of soil-related indices was derived, such as the brightness index (BI), salinity index (SI), color index (CI), dry soil index (DSI), and dry vegetation index (DVI). These indices were computed using established spectral band formulas described in prior literature [13], [51], [52], [53].

Furthermore, the TCT was applied to extract three orthogonal components—brightness, greenness, and wetness—which summarize dominant spectral trends in the landscape. TCT is particularly effective for capturing joint soil–vegetation dynamics and improving the interpretability of multispectral data. The mathematical formulation and implementation details of TCT are described in [54].

The final feature set integrates raw reflectance values from L8 with the derived VIs, SIs, and TCT components, yielding a comprehensive and biophysically informed representation of the surface environment. This enriched dataset provides a robust foundation for learning-based SOC estimation across heterogeneous land cover conditions.

By combining raw spectral bands with derived indices and TCT components, the feature space captures both direct reflectance information and higher-level biophysical descriptors, which have been shown to improve model robustness and generalization in heterogeneous soil–vegetation conditions.

D. Paired Dataset Preparation for ReflectGAN

To train ReflectGAN, a paired dataset of vegetated and corresponding bare-soil reflectance samples was constructed. Vegetated samples were selected with NDVI >0.2 to represent significant vegetation cover, while bare-soil samples were identified with NDVI <0.2 to ensure minimal vegetation influence. Each vegetated sample was paired with its closest bare-soil counterpart by minimizing the Euclidean distance between their geographic coordinates, under the assumption that SOC levels do not vary abruptly within small spatial

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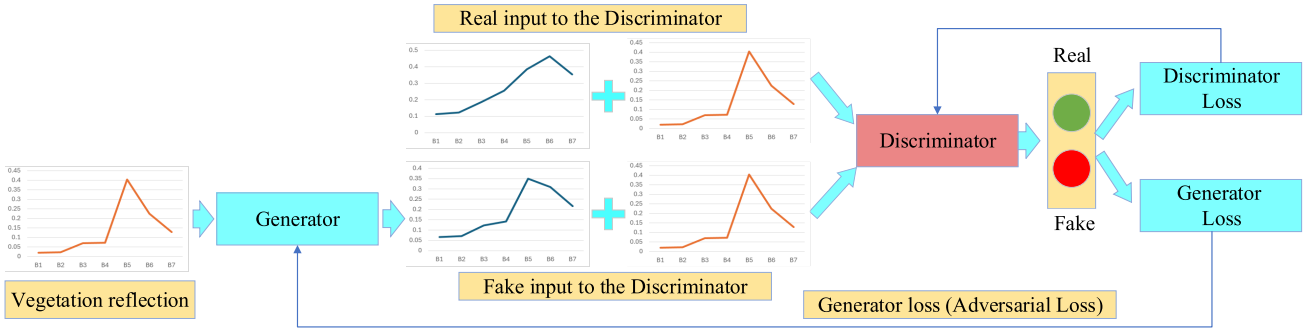


Fig. 1. Proposed ReflectGAN architecture illustrating the interaction between the generator and the discriminator. The generator transforms vegetated soil reflectance into bare-soil-like spectra, while the discriminator evaluates their authenticity.

extents [55]. This matching ensures that paired samples share similar soil properties apart from the influence of vegetation cover.

In cases where multiple bare-soil samples were found in close proximity to a vegetated sample, we computed the average reflectance of the nearest candidates to serve as a robust reference. This averaging approach reduces the influence of local noise or outliers and ensures that the resulting bare-soil target accurately represents typical reflectance in the surrounding area.

The final paired dataset consists of input–target reflectance pairs, where the input corresponds to the multispectral reflectance of a vegetated soil sample and the target is the aggregated bare-soil reflectance. These pairs were used to train ReflectGAN in a supervised learning setting, enabling the model to learn the structured transformation from vegetation-contaminated observations to vegetation-suppressed soil reflectance.

III. Proposed ReflectGAN

A. Overview and Design Principles of ReflectGAN

ReflectGAN is formulated as a conditional GAN designed to reconstruct bare-soil reflectance from vegetation-contaminated satellite observations for improved SOC estimation. The proposed framework is motivated by the need to recover soil-dominant spectral information that is obscured by vegetation cover in mixed land-cover conditions.

ReflectGAN integrates two key design principles: 1) spectral conditioning, whereby vegetated reflectance explicitly guides the generation process to preserve physical and spectral consistency, and 2) residual learning within the generator to enable stable and effective modeling of complex, nonlinear spectral transformations. Together, these components allow the model to suppress vegetation-induced distortions while retaining soil-relevant spectral characteristics.

The remainder of this section describes the adversarial formulation, loss functions, and detailed network architectures of ReflectGAN.

B. Overall Architecture of ReflectGAN

ReflectGAN is a conditional GAN designed to reconstruct bare-soil reflectance from vegetation-contaminated satellite observations to improve soil organic carbon (SOC) estimation under mixed land cover conditions. The model learns a direct spectral transformation from vegetated to bare-soil reflectance using paired observations, enabling effective suppression of vegetation-induced spectral interference while preserving soil-relevant information.

The proposed ReflectGAN consists of two primary components: a generator and a discriminator, which interact within an adversarial learning framework illustrated in Fig. 1. The generator aims to translate vegetation-contaminated soil reflectance into realistic bare-soil spectral signatures, while the discriminator is trained to distinguish real bare-soil reflectance from the generator's synthetic outputs. Through this adversarial process, the generator progressively improves its ability to reconstruct soil-specific spectral patterns from vegetation-affected inputs.

Both the generator and the discriminator are conditioned on the input-vegetated reflectance, ensuring that the generated outputs remain contextually aligned with observed spectral characteristics. This conditional formulation allows ReflectGAN to focus on recovering soil-dominant spectral information rather than generating unconstrained synthetic spectra.

ReflectGAN adopts the standard adversarial learning objective, defined as

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}\left[\log D(x)\right] + \mathbb{E}_{z\sim p_z(z)}\left[\log\left(1 - D(G(z))\right)\right] \quad (1)$$

where x denotes real samples drawn from the true data distribution $p_{\text{data}}(x)$, and z represents a noise vector sampled from a prior distribution $p_z(z)$. The discriminator D is trained to distinguish real from generated samples, while the generator G aims to produce synthetic outputs that are indistinguishable from real data.

To incorporate vegetation information explicitly, ReflectGAN extends this formulation to a conditional adversarial objective

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}\left[\log D(x|y)\right] + \mathbb{E}_{z\sim p_z(z)}\left[\log\left(1 - D(G(z|y))\right)\right] \quad (2)$$

where y represents the vegetated soil reflectance used as the conditioning variable. In this formulation, the generator

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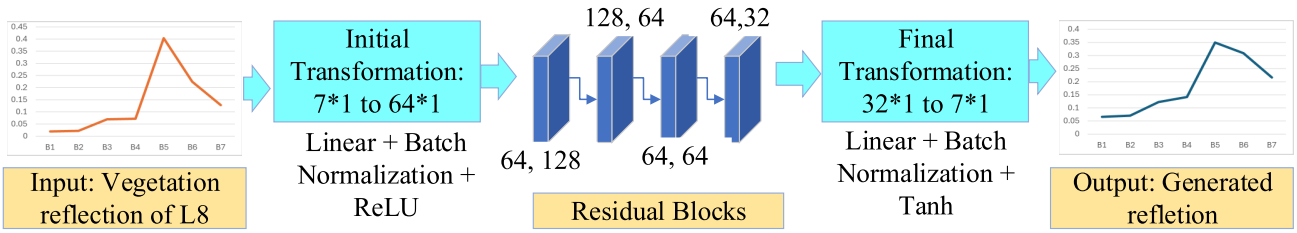


Fig. 2. Schematic of the proposed ReflectGAN generator, illustrating the transformation from input vegetated soil spectral signatures through a sequence of residual blocks to the reconstructed bare-soil spectra.

produces synthetic bare-soil spectra $G(z|y)$ guided by the input vegetation-affected reflectance, and the discriminator evaluates their realism given the same conditioning information.

This conditional adversarial framework enables ReflectGAN to generate spectrally realistic and context-consistent bare-soil reflectance, forming the basis for accurate SOC estimation in vegetated landscapes. Detailed descriptions of the loss functions and internal network architectures are provided in Sections III-C and III-D.

C. Calculation of Loss in ReflectGAN

In the ReflectGAN framework, adversarial training between the generator (G) and the discriminator (D) is governed by well-defined loss functions that guide the model toward accurately reconstructing bare-soil reflectance from vegetation-contaminated inputs. These loss functions ensure both the generator's ability to synthesize realistic spectra and the discriminator's capacity to distinguish real from synthetic data, based on inputs.

1) Discriminator Loss: The discriminator is trained to differentiate between real bare-soil spectral signatures and those generated by the generator, both conditioned on vegetated soil inputs. Its loss function consists of two components:

Real Data Loss ($\text{Loss}_{D_{\text{real}}}$): This term penalizes incorrect classification of real bare-soil data. Ideally, D should output values close to 1 for real samples

$$\text{Loss}_{D_{\text{real}}} = -\frac{1}{N}\sum \log\left(D\left(x|y\right)\right). \tag{3}$$

Generated Data Loss ($\text{Loss}_{D_{\text{fake}}}$): This term penalizes the discriminator when it incorrectly classifies synthetic data as real. The generator's output $G(z|y)$ is expected to be classified close to 0

$$\text{Loss}_{D_{\text{fake}}} = -\frac{1}{N}\sum \log\left(1 - D\left(G\left(z|y\right)\right)\right). \tag{4}$$

The total discriminator loss is the sum of both components

$$\text{Loss}_D = \text{Loss}_{D_{\text{real}}} + \text{Loss}_{D_{\text{fake}}}. \tag{5}$$

2) Generator Loss: The generator is trained to fool the discriminator by generating bare-soil reflectance that appears real. Its loss encourages D to classify synthetic data as authentic

$$\text{Loss}_G = -\frac{1}{N}\sum \log\left(D\left(G\left(z|y\right)\right)\right). \tag{6}$$

Here, x denotes the real bare-soil reflectance, y represents the vegetated soil input (the conditioning variable), z is the noise vector, and N is the batch size.

By incorporating the conditioning variable y into both networks, ReflectGAN ensures that synthetic outputs are not only realistic but also contextually aligned with the input vegetated reflectance. This conditional structure significantly enhances the model's ability to recover soil-specific signals and suppress vegetation-induced noise.

In summary, these adversarial losses enable ReflectGAN to learn complex, nonlinear mappings between vegetated and bare-soil spectral signatures. Iterative optimization of Loss_D and Loss_G drives the generator toward producing physically meaningful, high-fidelity reflectance outputs, ultimately improving SOC estimation from satellite imagery.

D. Detailed Architecture Overview

Building upon the previously outlined framework, this section provides a detailed description of the internal architectures of the generator and discriminator in ReflectGAN. The design choices are motivated by the need to accurately reconstruct realistic and environmentally consistent bare-soil spectral signatures from vegetation-contaminated inputs.

1) Generator: The generator is designed to transform vegetation-contaminated soil spectral signatures into bare-soil reflectance using Landsat 8 (L8) imagery. It adopts a deep feedforward architecture composed of multiple residual blocks, enabling effective feature learning and stable spectral translation. An overview of the generator structure is shown in Fig. 2.

Initial Linear Transformation: The input reflectance vector, consisting of the seven L8 spectral bands, is first projected into a higher-dimensional feature space. Specifically, a fully connected layer maps the 7-D input to 64 feature channels, followed by batch normalization and a rectified linear unit (ReLU) activation to introduce nonlinearity and stabilize training

$$x_1 = \text{ReLU}\left(\text{BN}\left(W_0 x + b_0\right)\right) \tag{7}$$

where $W_0 \in \mathbb{R}^{64\times 7}$ and $b_0 \in \mathbb{R}^{64}$ denote the weights and biases, respectively.

Residual Blocks: The core of the generator consists of a sequence of residual blocks. Each block applies two linear transformations with batch normalization and ReLU activation, followed by an identity skip connection that adds the input of

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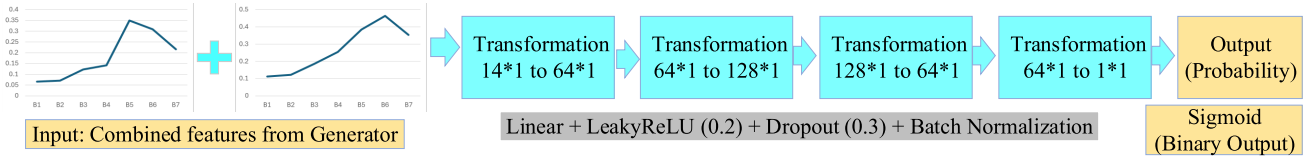


Fig. 3. Schematic of the proposed ReflectGAN discriminator, showing the processing pipeline from concatenated vegetated and bare-soil spectral inputs to the final binary classification output.

the block to its output. This structure facilitates deeper feature learning while preserving essential spectral information. The residual mapping is defined as

$$x_{i+1} = x_i + \mathrm{ReLU}\left(\mathrm{BN}\left(W_{i1} \cdot \mathrm{ReLU}\left(\mathrm{BN}\left(W_{i0}x_i + b_{i0}\right)\right) + b_{i1}\right)\right) \tag{8}$$

where W_{i0}, W_{i1} and b_{i0}, b_{i1} are the parameters of the ith residual block.

The feature dimensionality is progressively adjusted throughout the network. The architecture initially expands the feature space from 64 to 128 dimensions to enhance representational capacity, then gradually compresses it to 64 and finally to 32 dimensions. This expansion–compression strategy balances expressive power and computational efficiency.

Output Layer: After passing through the residual blocks, a final linear layer maps the 32-D feature vector back to the original 7-D spectral space. A tanh activation function is applied to constrain the output reflectance values within a bounded range

$$G(x) = \tanh\left(W_f x_N + b_f\right) \tag{9}$$

where $W_f \in \mathbb{R}^{7\times 32}$ and $b_f \in \mathbb{R}^7$ denote the output layer parameters.

Through hierarchical feature extraction and residual learning, the generator effectively models complex nonlinear transformations required to reconstruct bare-soil reflectance from vegetation-affected inputs.

2) Discriminator: The discriminator serves as a binary classifier within the adversarial learning framework, distinguishing real bare-soil spectra from those generated by the generator. Its architecture is illustrated in Fig. 3.

Input Processing: The discriminator receives two spectral inputs: a vegetated soil reflectance vector and a corresponding bare-soil reflectance vector (either real or generated). These two 7-D vectors are concatenated to form a 14-D input

$$z = \mathrm{concat}(x, y) \tag{10}$$

where x denotes the vegetated soil spectrum and y denotes the bare-soil spectrum.

Feature Extraction and Classification: The concatenated input is processed through a series of fully connected layers with LeakyReLU activations, batch normalization, and dropout regularization. The first layer expands the feature space to 64 dimensions

$$z_1 = W_1 z + b_1 \tag{11}$$

followed by a LeakyReLU activation with a negative slope of 0.2 and a dropout layer with a rate of 0.3. Additional layers further refine the feature representation, expanding to 128 dimensions and subsequently compressing back to 64 dimensions.

Output Layer: The final layer reduces the feature vector to a single scalar, which is passed through a sigmoid activation to produce a probability score indicating whether the input bare-soil spectrum is real or generated

$$D(x, y) = \sigma\left(W_f z_N + b_f\right) \tag{12}$$

where $\sigma(\cdot)$ denotes the sigmoid function.

By jointly analyzing vegetated and bare-soil spectra, the discriminator enforces consistency between input conditions and generated outputs, thereby strengthening adversarial training and improving the realism of reconstructed soil reflectance.

E. Physical Interpretation and Parameter Robustness of ReflectGAN

Although ReflectGAN is implemented as a data-driven DL framework, its design is grounded in physically meaningful assumptions about soil–vegetation spectral interactions. In mixed pixels, the observed reflectance represents a nonlinear combination of soil background reflectance, vegetation scattering, and multiple interaction effects driven by canopy structure, illumination geometry, and wavelength-dependent absorption. Traditional spectral-domain approaches, such as VIs or linear spectral unmixing, rely on simplified assumptions (e.g., linear mixing, fixed endmembers, or empirical thresholds) that often fail under moderate to dense vegetation cover.

ReflectGAN improves upon these approaches by learning a conditional vegetation-to-soil spectral transformation directly in reflectance space. By conditioning the generator on vegetation-contaminated reflectance, the model implicitly preserves physically consistent spectral dependencies present in the observed signal while suppressing vegetation-dominated contributions. The use of residual learning further reinforces this physical interpretation: residual blocks encourage the network to learn corrective spectral adjustments rather than unconstrained transformations, thereby retaining soil-relevant spectral structure while progressively removing vegetation-induced distortions. This formulation aligns with the physical expectation that vegetation effects act as structured perturbations on underlying soil reflectance rather than entirely replacing it.

In contrast to spectral unmixing methods that estimate abundance fractions, ReflectGAN explicitly reconstructs bare-soil reflectance, enabling direct compatibility with conventional SOC estimation models that operate on spectral inputs. This

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reflectance-centric strategy allows ReflectGAN to capture nonlinear soil–vegetation interactions without requiring predefined endmembers or explicit physical parameterization, which are often unavailable or unstable across heterogeneous landscapes.

Regarding parameter robustness, ReflectGAN is designed with stable architectural and optimization choices that reduce sensitivity to dataset-specific tuning. Key hyperparameters, including the learning rate and the number of residual blocks, were selected based on convergence stability and reconstruction fidelity rather than region-specific characteristics. Within a stable range, moderate adjustments to the learning rate primarily affect convergence speed rather than reconstruction quality, while the depth of the residual structure controls the model's capacity to represent spectral complexity. Datasets exhibiting higher spectral variability or denser vegetation cover may benefit from deeper residual architectures, whereas simpler landscapes can be modeled effectively with fewer blocks without loss of performance.

Importantly, the conditional formulation constrains the solution space by anchoring generation to observed reflectance, which improves generalization across regions and sensors. As demonstrated by consistent performance across L8 and S2 data, ReflectGAN exhibits robustness to differences in spectral response functions and spatial resolution. For deployment in new regions, we recommend maintaining the same architectural structure while performing limited validation-based tuning of the learning rate and training epochs, rather than extensive hyperparameter re-optimization. This design philosophy supports practical transferability while preserving physical interpretability and methodological stability.

F. Key Contributions of the ReflectGAN

The ReflectGAN model brings several key innovations to the application of GANs in environmental remote sensing, particularly for transforming vegetation-contaminated satellite data into accurate bare-soil reflections. Its design and implementation contribute meaningfully to both theoretical advancements and practical applications in remote sensing and land monitoring.

1) *Conditional Spectral Translation*: By conditioning on vegetated soil spectral signatures, ReflectGAN ensures that generated bare-soil outputs are not only realistic but also contextually relevant, addressing a major gap in traditional GAN applications for environmental data.
2) *Advanced Feature Learning with Residual Blocks*: The integration of residual learning within the generator enables the model to capture complex, nonlinear spectral transformations, while also mitigating challenges like vanishing gradients, leading to more stable and efficient training.
3) *Robust Discriminative Analysis*: The tailored discriminator architecture critically evaluates concatenated vegetated and bare-soil spectra, enhancing adversarial training dynamics and improving the realism of the generated soil reflections.
4) *Enhanced Environmental Monitoring*: ReflectGAN offers a novel approach for generating accurate soil reflectance from vegetated areas, supporting more reliable SOC estimation, soil health assessment, and agricultural monitoring from satellite imagery.
5) *Broader Applicability*: Although designed for remote sensing, the conditional learning and spectral translation strategy of ReflectGAN can be adapted to other domains where disentangling complex mixed signals is necessary, such as urban mapping, forestry, and climate change monitoring.

Overall, ReflectGAN represents a significant advancement in the application of GANs for environmental remote sensing, combining architectural innovation with practical relevance to address longstanding challenges in soil analysis from satellite imagery.

IV. Proposed Method

This study introduces a ReflectGAN-based framework to improve SOC estimation from L8 imagery by mitigating the impact of vegetation on spectral reflectance. The overall pipeline, illustrated in Fig. 4, begins with remote sensing data acquisition and ends with SOC estimation using both ML and DL models.

L8 imagery was preprocessed to ensure spectral consistency and spatial accuracy. This involved radiometric and atmospheric correction, followed by pansharpening using the 15-m panchromatic band to enhance the resolution of the 30-m multispectral bands. These steps converted raw data into surface reflectance values and enabled the precise extraction of spectral signatures at soil sampling locations.

To address vegetation interference, we included both bare and vegetated soil samples, identified using the NDVI. By covering NDVI values across a broader range, the study evaluates model robustness under varying vegetation conditions, especially in scenarios where vegetation masks underlying soil properties.

To ensure a fair and consistent comparison, all baseline ML and DL models were optimized using the same grid-search strategy and identical training–validation splits. Hyperparameter search ranges were selected based on commonly adopted values in SOC estimation literature, rather than exhaustive dataset-specific tuning, in order to reduce overfitting and improve generalization. Importantly, ReflectGAN hyperparameters were selected based on reconstruction stability and convergence behavior and were not optimized using SOC labels, whereas all regression models were tuned directly for SOC prediction. This design ensures that performance gains primarily reflect the effectiveness of vegetation correction rather than differences in hyperparameter optimization effort.

A core innovation in this study is the ReflectGAN model, which is specifically designed to reconstruct soil-specific reflectance from vegetation-affected L8 samples. Unlike generic image-to-image translation GANs, ReflectGAN leverages paired vegetated and bare-soil samples to learn a conditional mapping that suppresses vegetation influence while preserving soil-related spectral characteristics. The generator is trained to reconstruct clean reflectance patterns that resemble bare soil, guided by a discriminator that enforces spectral realism and consistency. This design ensures that the

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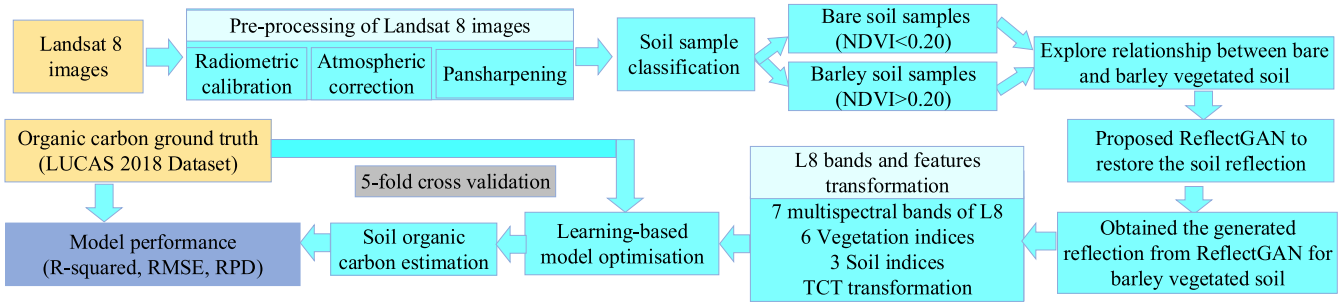


Fig. 4. Illustration of the proposed framework for SOC estimation, encompassing data preprocessing, ReflectGAN-based reflectance correction, model training, and evaluation.

TABLE II
SUMMARY OF OPTIMIZED HYPERPARAMETER CONFIGURATIONS FOR EACH MODEL USING GRID SEARCH

Model	Tuned Hyperparameters
LR	Standardization: True; Random State: 42
RF	Number of Estimators: 100; Random State: 42
DT	Random State: 42; Max Depth: None
GB	Number of Estimators: 100; Random State: 42
CBR	Learning Rate: 0.1; Tree Depth: 10; Loss Function: RMSE; Number of Iterations: 100
KNN	Number of Neighbors: 5; Distance Metric: Euclidean
ANN	Hidden Layers: 3 (Units: 128-64-1); Activation: ReLU; Dropout Rate: 0.3; Batch Size: 32; Epochs: 100; Learning Rate: 0.001
SVR	Kernel: RBF; Regularization (C): 1.0; Epsilon: 0.2; Batch Size: 32; Epochs: 100; Learning Rate: 0.001
1dCNN	Convolutional Layers: 3; Filters: 16-32-64; Kernel Size: 3; Padding: 1; Fully Connected Layer: 128 units; Dropout Rate: 0.3; Batch Size: 32; Epochs: 100; Learning Rate: 0.001

recovered reflectance remains meaningful for downstream SOC estimation.

In addition to L8 reflectance, auxiliary features were incorporated to enhance the dataset. These included VIs (e.g., NDVI, SAVI, and MSAVI), SIs (e.g., BI, SI, and DSI), and TCT components (brightness, greenness, and wetness). Derived from the L8 bands, these features provide additional context on vegetation density and soil conditions relevant to SOC distribution.

The enriched dataset was used to train a range of ML and DL models, including linear regression (LR), random forest (RF), decision tree (DT), gradient boosting (GB), catBoost regressor (CBR), K-nearest neighbors (KNNs), support vector regression (SVR), artificial neural networks (ANNs), and a 1-D convolutional neural network (1D-CNN). Each model was systematically optimized through grid search, and the final hyperparameter settings are summarized in Table II. Model performance was assessed using standard evaluation metrics: coefficient of determination (R^2), root mean squared error (RMSE), and residual prediction deviation (RPD).

These metrics are defined as follows:

$$R^2 = 1 - \frac{\sum (y_{\text{true}} - \hat{y}_{\text{est}})^2}{\sum (y_{\text{true}} - \bar{y}_{\text{avg}})^2} \tag{13}$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_{i,\text{true}} - \hat{y}_{i,\text{est}})^2} \tag{14}$$

$$RPD = \frac{\text{std_dev}}{RMSE} \tag{15}$$

where y_{true} and $\hat{y}_{\text{est}}$ represent the actual and estimated SOC values, $\bar{y}_{\text{avg}}$ is the mean of the actual values, N is the total number of observations, and std_dev is the standard deviation of the observed values.

These metrics collectively offer insight into both estimation accuracy and generalization capability. The ReflectGAN-corrected reflectance, when used as input to these models, aims to reduce spectral noise and recover underlying soil properties, especially in vegetation-dominated areas, thereby improving SOC estimation performance.

V. RESULTS

This section evaluates the effectiveness of ReflectGAN in enhancing SOC estimation under mixed bare and vegetated conditions. We first examine the correlation between spectral bands and SOC across different input types, followed by a performance comparison of ML and DL models using various reflectance scenarios.

Table III presents Pearson's correlation coefficients between L8 spectral bands (B1–B7) and SOC for three different input types: bare-soil, vegetated soil, and ReflectGAN-reconstructed reflectance. As expected, bare-soil reflectance exhibited stronger negative correlations with SOC, particularly in the shortwave infrared bands (B6 and B7; −0.44 and −0.46, respectively), reaffirming their importance in SOC estimation. In contrast, vegetated samples showed substantially weaker correlations, highlighting the masking effect of vegetation on soil reflectance. Notably, ReflectGAN-reconstructed soil reflectance demonstrated the strongest correlations across all bands, especially within the red-edge and shortwave infrared regions (B4 to B7), reaching up to −0.74.

These results indicate that ReflectGAN effectively mitigates vegetation-induced distortion and enhances the fidelity of soil reflectance, providing more reliable inputs for SOC modeling. The similarity between bare-soil reflectance and ReflectGAN-reconstructed reflectance, both in terms of correlation values and visual patterns (Fig. 5), indicates that ReflectGAN preserves the physical integrity of bare-soil spectra even when reconstructed from vegetation-contaminated observations.

Table IV summarizes SOC estimation results across different reflectance input types. The bare-soil scenario yielded strong baseline performance, with RF achieving the best

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TABLE III

PEARSON'S CORRELATION COEFFICIENTS BETWEEN LANDSAT 8 SPECTRAL BANDS (B1–B7) AND SOC FOR DIFFERENT INPUT TYPES, ILLUSTRATING HOW REFLECTGAN IMPROVES SPECTRAL CORRELATION COMPARED TO VEGETATED SOIL INPUTS

Input Type	B1	B2	B3	B4	B5	B6	B7
Bare Soil Only	-0.27	-0.29	-0.40	-0.36	-0.34	-0.44	-0.46
Vegetated Soil Only	-0.08	-0.10	-0.14	-0.14	-0.04	-0.13	-0.15
ReflectGAN-Reconstructed Soil (Proposed)	-0.43	-0.49	-0.65	-0.68	-0.67	-0.73	-0.74

TABLE IV

SOC ESTIMATION PERFORMANCE ($R^2\uparrow$, $RMSE\downarrow$, $RPD\uparrow$) COMPARISON OF ESTABLISHED ML AND DL MODELS USING DIFFERENT REFLECTANCE INPUTS, INCLUDING REFLECTGAN-RECONSTRUCTED REFLECTANCE. ALL BASELINE MODELS WERE OPTIMIZED USING A UNIFIED GRID-SEARCH STRATEGY WITH IDENTICAL TRAINING–VALIDATION SPLITS, WHILE REFLECTGAN EMPLOYED A FIXED ARCHITECTURE WITH STABLE TRAINING PARAMETERS. RESULTS WERE EVALUATED UNDER THE SAME CROSS-VALIDATION PROTOCOL

Input	LR	RF	DT	GB	CBR	KNN	ANN	SVR	1D-CNN
Bare Soil Only	0.28, 9.11, 1.19	0.65, 6.41, 1.71	0.43, 7.98, 1.39	0.58, 6.87, 1.59	0.60, 6.74, 1.62	0.53, 7.27, 1.49	0.41, 8.26, 1.31	0.39, 8.50, 1.28	0.50, 7.45, 1.45
Bare Soil With Additional Features	0.29, 9.02, 1.21	0.61, 0.71, 1.61	0.31, 8.87, 1.27	0.55, 7.14, 1.52	0.64, 6.30, 1.72	0.53, 7.04, 1.54	0.44, 7.72, 1.41	0.40, 8.41, 1.29	0.51, 6.95, 1.58
Vegetated Soil Only	0.00, 6.66, 0.99	-0.09, 6.92, 0.95	-1.01, 9.34, 0.71	-0.04, 6.76, 0.98	-0.03, 6.74, 0.98	-0.16, 7.14, 0.92	-0.01, 6.67, 0.99	-0.03, 6.75, 0.98	-0.06, 6.85, 0.97
Vegetated Soil With Additional Features	-0.04, 6.79, 0.98	-0.09, 6.91, 0.96	-0.03, 9.39, 0.70	-0.04, 6.76, 0.98	-0.01, 6.68, 0.99	-0.22, 7.31, 0.90	-0.06, 6.83, 0.97	-0.03, 6.72, 0.98	-0.18, 7.23, 0.92
Bare Soil + Raw Vegetated	0.17, 8.69, 1.10	0.55, 6.32, 1.51	0.25, 8.14, 1.18	0.47, 6.89, 1.38	0.51, 6.65, 1.44	0.39, 7.41, 1.29	0.34, 7.74, 1.23	0.22, 8.44, 1.13	0.47, 6.83, 1.41
Bare Soil + Raw Vegetated With Additional Features	0.15, 8.77, 1.09	0.52, 6.51, 1.47	0.24, 8.13, 1.19	0.48, 6.85, 1.39	0.50, 6.76, 1.42	0.39, 7.43, 1.29	0.34, 7.71, 1.24	0.25, 8.24, 1.16	0.40, 7.31, 1.31
ReflectGAN-Reconstructed Soil (Proposed)	0.50, 4.14, 1.95	0.53, 3.93, 2.11	0.27, 5.39, 1.35	0.50, 4.24, 1.82	0.53, 3.95, 2.10	0.47, 4.27, 1.88	0.50, 4.09, 2.02	0.54, 3.95, 2.07	0.51, 4.07, 2.01
ReflectGAN-Reconstructed Soil With Additional Features	0.50, 4.18, 1.91	0.51, 4.14, 1.94	0.23, 5.55, 1.30	0.51, 4.23, 1.82	0.53, 4.05, 1.96	0.46, 4.40, 1.78	0.49, 4.14, 1.98	0.50, 4.23, 1.86	0.47, 4.33, 1.83
Bare Soil + ReflectGAN-Reconstructed Soil	0.36, 7.57, 1.27	0.65, 5.50, 1.75	0.41, 7.23, 1.33	0.60, 5.95, 1.60	0.64, 5.59, 1.70	0.55, 6.27, 1.52	0.49, 6.76, 1.42	0.43, 7.17, 1.33	0.50, 6.65, 1.44
Bare Soil + ReflectGAN-Reconstructed Soil With Additional Features	0.39, 7.49, 1.29	0.61, 5.84, 1.64	0.52, 6.49, 1.47	0.56, 6.21, 1.53	0.60, 5.96, 1.61	0.53, 6.38, 1.51	0.54, 6.36, 1.51	0.41, 7.30, 1.31	0.47, 6.62, 1.49

Note: ReflectGAN-reconstructed soil refers to reflectance reconstructed from vegetated observations using a paired GAN model specifically trained to minimize vegetation interference.

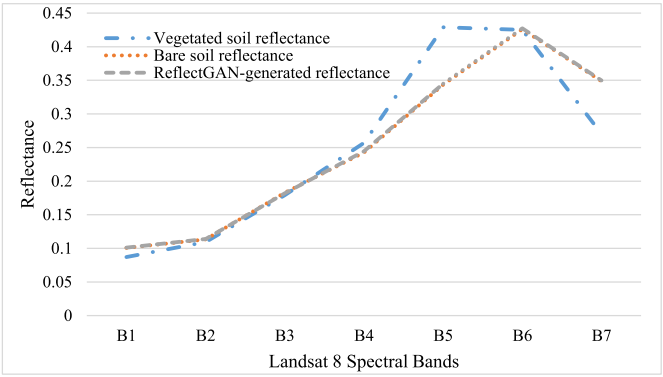
results (R^2 = 0.65, $RMSE$ = 6.41, RPD = 1.71), likely due to the absence of vegetation interference (see Table IV). Incorporating additional features (such as VIs, SIs, and TCT components) slightly improved performance for some models (e.g., CBR and 1dCNN), suggesting that supplementary spectral information can enhance learning in specific cases. However, for other models, including extra features, led to comparable or marginally reduced performance, indicating that when vegetation effects are minimal, the benefit of additional indices remains limited.

Models trained on vegetated soil inputs exhibited poor performance across all approaches due to the dominance of plant reflectance, with several models yielding negative or near-zero R^2 values (as shown in Table IV). The inclusion of additional features did not substantially improve these results. These findings confirm that vegetation contamination severely masks soil spectral information and that conventional feature augmentation alone is insufficient to recover meaningful SOC-related patterns from vegetation-affected observations.

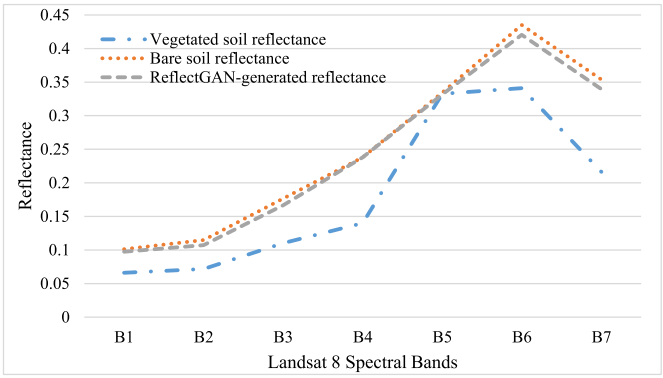
Combining vegetated and bare-soil samples led to slight improvements compared to using vegetated samples alone, as the models could partially leverage information from bare-soil observations. The addition of further features, such as vegetation and SIs, resulted in only minor gains for a few models, while overall performance remained largely unchanged. Despite these adjustments, the results continued to lag behind those achieved using bare-soil data alone, underscoring that vegetation contamination fundamentally limits the reliability of SOC estimation.

ReflectGAN, which reconstructs bare-soil reflectance from vegetation-contaminated observations using a paired-GAN architecture, substantially improved SOC estimation performance across all models, as shown in Table IV. The best result was achieved by RF, with an R^2 of 0.53, $RMSE$ of 3.93, and RPD of 2.11, indicating strong accuracy and reliability. Other models, including LR, CBR, ANN, and 1dCNN, also achieved RPD values exceeding 2.0, demonstrating that ReflectGAN consistently enhances SOC estimation across diverse ML and DL algorithms. This general applicability underscores the robustness of ReflectGAN as a versatile solution for SOC modeling. The addition of extra features did not yield further improvements, indicating that ReflectGAN-reconstructed

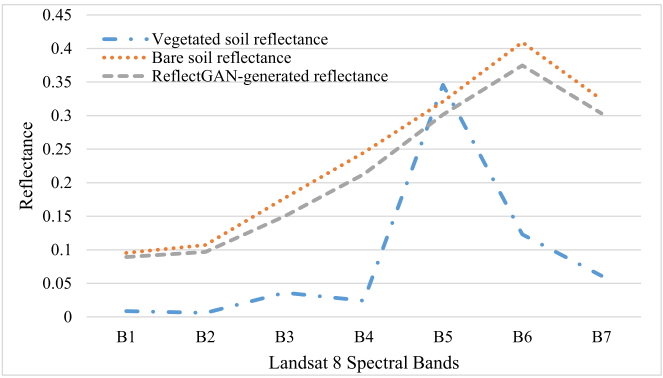
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(a)



(b)



(c)

Fig. 5. Comparison of different soil reflectance types against ReflectGAN-reconstructed reflectance, illustrating how the proposed method transforms vegetated reflectance into bare-soil reflectance under varying vegetation densities. (a) Lightly vegetated soil reflectance (NDVI = 0.25). (b) Moderately vegetated soil reflectance (NDVI = 0.42). (c) Highly vegetated soil reflectance (NDVI = 0.83).

reflectance alone provides sufficient information for accurate SOC estimation. This behavior can be explained by the fact that ReflectGAN explicitly reconstructs soil-dominant spectral signatures by suppressing vegetation-induced distortions, thereby recovering the core spectral information required for SOC estimation. Many auxiliary features, such as VIs and TCT components, are derived from nonlinear combinations of the original spectral bands and are primarily designed to enhance soil–vegetation separability. Once vegetation effects are effectively mitigated through reflectance reconstruction, these derived features provide limited additional information and may introduce redundancy, leading to marginal or inconsistent performance gains.

Combining real bare-soil samples with ReflectGAN-reconstructed reflectance resulted in the highest R^2 values across several models, with RF achieving the best R^2 of 0.65 (see Table IV). However, RMSE and RPD values were not consistently optimal compared to using ReflectGAN-reconstructed data alone, likely due to distributional differences between real and generated reflectance. The addition of extra features led to minor improvements, particularly in models such as LR, DT, and ANN, while most other models showed comparable or slightly reduced performance. These results suggest that although hybrid training helps capture broader spectral variability, it may also introduce additional complexity that limits further gains in estimation reliability.

The results demonstrate that models trained on bare-soil reflectance consistently achieve superior SOC estimation performance compared to those using vegetated inputs. ReflectGAN-reconstructed reflectance further enhanced model accuracy by minimizing vegetation-induced distortions, enabling reliable SOC estimation across mixed land cover conditions. These findings highlight the potential of reflectance reconstruction strategies to improve soil monitoring across heterogeneous landscapes.

VI. Discussion

This section explores the performance and broader significance of the proposed ReflectGAN framework for SOC estimation. We first compare ReflectGAN against existing vegetation correction methods, then evaluate its generalizability across different satellite datasets. Finally, we discuss its practical strengths and limitations, propose future research directions, and highlight the broader implications for operational soil monitoring and remote sensing applications.

A. Comparison With Existing Vegetation Correction Methods

We evaluate the performance of the proposed ReflectGAN approach against a range of existing vegetation correction methods, encompassing both traditional techniques and advanced DL-based models from recent literature. The selected models reflect widely used or representative approaches in the literature that address vegetation interference via index-based filtering, physical modeling, spectral unmixing (linear and probabilistic), and GAN-based translation. While more recent nonlinear GAN-based unmixing models, such as HyperGAN [41] and 3D-GAN [44], have shown promising results in hyperspectral analysis, they are primarily designed for abundance estimation and spectral classification rather than direct reflectance reconstruction, which is the primary requirement for SOC estimation in this study.

In particular, nonlinear unmixing GANs aim to decompose mixed spectra into endmember abundance fractions, which are well-suited for material composition analysis but are not directly compatible with conventional SOC regression models that operate on spectral reflectance inputs. As a result, these methods address a different problem formulation than that considered in this study.

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TABLE V
PERFORMANCE COMPARISON OF THE PROPOSED REFLECTGAN METHOD AGAINST EXISTING VEGETATION CORRECTION APPROACHES FOR SOC ESTIMATION USING THE BEST-PERFORMING RF MODEL, WITH UNIFORM HYPERPARAMETERS APPLIED ACROSS ALL METHODS TO ENSURE FAIR EVALUATION

Approach	Vegetation Handling	R^2	RMSE	RPD
Bare Soil Only	No Vegetation Correction (Bare Soil Samples Only)	0.65	6.41	1.71
Vegetation Index-Based Correction [19]	Partial Correction (NDVI, SAVI)	-0.09	6.90	0.95
Spectral Mixture Analysis (SMA) [22], [23]	Linear Spectral Mixing Model	0.21	8.25	1.14
Adaptive Spectral Mixture Analysis (ASMA) [24]	Adaptive Linear Spectral Unmixing	0.11	6.38	1.06
Radiative Transfer Model (PROSAIL) [21]	Physically Based Model Correction	0.24	8.13	1.19
Probabilistic Mixture Model-Based Spectral Unmixing (PMM-SU) [28], [29]	Bayesian Spectral Unmixing	0.40	6.95	1.45
Pix2Pix GAN [33]	Paired Image-to-Image Translation	0.16	6.16	1.09
CycleGAN [34]	Unpaired Image-to-Image Translation	0.08	7.10	1.02
ReflectGAN (Proposed)	**Soil-Specific GAN-Based Transformation**	**0.53**	**3.93**	**2.11**

The comparative results of all evaluated vegetation correction methods, summarized in Table V, provide insight into their relative effectiveness in supporting SOC estimation under vegetation-influenced conditions.

The RF model was chosen for this comparison as it consistently performed well in most cases (Table IV). Additionally, RF is a robust and widely used algorithm for SOC estimation due to its simplicity and effective handling of high-dimensional data.

Bare soil only serves as a baseline scenario where only samples with no or minimal vegetation cover were used. As expected, this approach achieved the highest performance among all conventional methods (R^2 = 0.65, RMSE = 6.41, RPD = 1.71), primarily due to the absence of vegetation interference, which enabled a direct relationship between soil reflectance and SOC to be captured (see Table V). However, despite its strong performance, this approach has limited practical scalability, as bare-soil pixels are infrequent and seasonally constrained across most agricultural and natural landscapes.

While bare-soil samples provide the most reliable inputs, many landscapes are dominated by partial or full vegetation cover, necessitating correction strategies. Vegetation index-based correction methods, such as NDVI and SAVI, were tested to mitigate vegetation influence; however, they showed poor performance (R^2 = –0.09, RMSE = 6.90, RPD = 0.95). These indices indirectly account for vegetation cover but fail to recover the true underlying soil reflectance. As a result, they are ineffective at disentangling mixed signals, particularly when vegetation density varies across samples.

SMA moderately improved performance (R^2 = 0.21, RMSE = 8.25, RPD = 1.14), indicating some success in linearly separating vegetation and soil components. However, it assumes fixed endmembers and linear mixing, assumptions that often do not hold under real-world conditions where nonlinear reflectance interactions are common, limiting its effectiveness.

ASMA achieved a slightly better RMSE than SMA but resulted in a lower R^2 (0.11), likely due to unstable estimation of endmember fractions. While ASMA adaptively adjusts spectral endmembers for each pixel, this flexibility can introduce noise, particularly under dense vegetation cover.

PROSAIL, a physically based correction model, yielded marginal improvements over SMA (R^2 = 0.24) but still exhibited a high RMSE. Although PROSAIL accurately simulates canopy–soil reflectance interactions, its reliance on numerous input parameters (e.g., LAI, leaf angle, and soil moisture) limits its practical applicability across large, heterogeneous landscapes.

PMM-SU achieved improved performance (R^2 = 0.40, RPD = 1.45) by leveraging Bayesian inference to robustly estimate soil and vegetation fractions. Its probabilistic framework accounts for uncertainty, offering better generalization than linear SMA approaches; however, it still struggles to recover true bare-soil reflectance under high vegetation densities.

Among DL-based approaches, Pix2Pix GAN demonstrated modest gains (R^2 = 0.16), but struggled to generalize beyond paired training data. Although it outperformed traditional unmixing methods, its outputs occasionally retained vegetation artifacts, limiting its effectiveness for accurate SOC estimation.

CycleGAN, trained without paired supervision, performed the worst among GAN-based models (R^2 = 0.08), likely due to misalignment between input and target domains. Without explicit one-to-one correspondence between vegetated and bare-soil samples, CycleGAN struggled to learn consistent spectral transformations under conditions of high spectral variability.

In contrast to abundance-oriented unmixing GANs and generic image-to-image translation models, ReflectGAN is explicitly formulated as a reflectance reconstruction framework. It does not require predefined endmembers, abundance constraints, or material fraction supervision. Instead, ReflectGAN learns a paired spectral transformation that suppresses vegetation-induced distortions while preserving soil-relevant

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TABLE VI
REGIONAL-SCALE SOC ESTIMATION PERFORMANCE BASED ON SPATIAL AGGREGATION OF TEST-FOLD PREDICTIONS WITHIN 1.0° × 1.0° GRID CELLS

Reflectance Representation	R^2	RMSE	RPD	Regions (avg.)
Vegetated Reflectance (B1–B7)	-0.10	6.87	0.97	5.6
ReflectGAN-Generated Bare Reflectance (Proposed)	**0.52**	**2.48**	**1.98**	**5.6**

TABLE VII
DESCRIPTIVE STATISTICAL PARAMETERS FOR THE SOC DATASET INVESTIGATED USING SENTINEL-2 DATA

Soil Type	NDVI Range	Sample No.	Min	Max	Mean	Median	Std. Dev.	CV(%)
Mixed	$0 < \text{NDVI} < 0.93$	229	2.7	81.4	17.35	14.80	10.43	60.17

spectral characteristics directly in reflectance space. This design enables seamless integration with conventional machine learning and DL SOC models, which typically operate on reflectance inputs rather than abundance representations.

Finally, ReflectGAN (Proposed) significantly outperformed all competing approaches (R^2 = 0.53, RMSE = 3.93, RPD = 2.11). By learning soil-specific spectral transformations from paired vegetated and bare-soil reflectance samples, ReflectGAN effectively suppresses vegetation-induced noise and reconstructs physically realistic soil signals. Its superior performance highlights the critical importance of domain-specialized generative models for addressing complex reflectance mixing challenges in SOC estimation.

To rigorously assess the differences among the vegetation correction approaches, we conducted a one-way analysis of variance (ANOVA) separately for each SOC estimation metric (R^2, RMSE, and RPD) based on cross-validation results. The ANOVA indicated statistically significant differences across methods for all three metrics ($p < 0.001$). Subsequently, Tukey's honest significant difference (HSD) post-hoc analysis was performed to identify specific pairwise comparisons. The results confirmed that ReflectGAN significantly outperformed all other approaches, exhibiting higher R^2 and RPD values and lower RMSE values across all comparisons (all pairwise $p < 0.01$). These findings provide strong statistical evidence supporting the superiority of ReflectGAN for SOC estimation relative to both traditional vegetation correction methods and baseline GAN architectures.

In summary, traditional vegetation correction techniques are often constrained by oversimplified assumptions and limited spectral separation capabilities. In contrast, GAN-based methods—particularly ReflectGAN—demonstrate strong potential for advancing SOC estimation by learning complex, nonlinear transformations that more accurately reconstruct true soil reflectance under vegetative conditions.

B. Regional-Scale Evaluation Under Spatial Heterogeneity

While Table V focuses on point-scale SOC estimation performance across different vegetation correction strategies, point-level accuracy does not necessarily imply strong applicability at the regional scale due to spatial heterogeneity in geographic and environmental conditions. To address this limitation, an additional regional-scale evaluation was conducted using georeferenced samples.

Specifically, SOC estimations obtained from the held-out test samples were spatially aggregated within fixed geographic grid cells of approximately 100×100 km (1.0°×1.0°). Samples falling within the same grid cell were treated as a single region, and the regional mean estimated SOC was compared against the regional mean observed SOC. Grid cells containing fewer than three samples were excluded to ensure stable regional statistics, resulting in approximately 5–6 regions per fold.

The resulting regional-scale performance is summarized in Table VI. As expected, SOC estimation based on vegetated reflectance remains unreliable even after spatial aggregation. In contrast, ReflectGAN-generated bare-soil-equivalent reflectance maintains strong regional-scale performance, demonstrating that the point-scale improvements achieved by the proposed method translate to regional-scale applicability under heterogeneous conditions.

C. Evaluating ReflectGAN Performance on Sentinel-2 Data

To validate the generalizability of ReflectGAN beyond L8 data, additional experiments were conducted using S2 multispectral imagery. The corresponding SOC dataset, summarized in Table VII, encompasses a wide range of SOC values (2.7–81.4 g/kg) and vegetation conditions (NDVI ranging from 0 to 0.93), providing a robust and diverse testbed for performance evaluation.

As shown in Table VIII, models trained exclusively on bare-soil samples achieved strong performance, with CBR yielding the highest results (R^2 = 0.64, RMSE = 4.78, RPD = 1.74). In contrast, models trained directly on vegetated soil samples without any correction exhibited severe performance degradation across all approaches, underscoring the significant spectral distortion introduced by vegetation cover. Simply merging bare and vegetated samples without correction also led to poor performance, further highlighting the challenges of mixed soil–vegetation reflectance.

Training models on ReflectGAN-corrected reflectance substantially improved SOC estimation. Among the tested models, RF achieved the best results using corrected data (R^2 = 0.55, RMSE = 5.35, RPD = 2.26), confirming the effectiveness of ReflectGAN in recovering soil-specific information from vegetated observations. Furthermore, combining real bare-soil samples with ReflectGAN-corrected data enhanced model robustness, with RF achieving strong estimation accuracy (R^2 = 0.62, RMSE = 5.56, RPD = 1.72).

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TABLE VIII
SOC ESTIMATION PERFORMANCE (R^2, $RMSE$, RPD) USING ML AND DL MODELS WITH SENTINEL-2 REFLECTANCE INPUTS

Input	LR	RF	DT	GB	CBR	KNN	ANN	SVR	1D-CNN
Bare Soil Only	0.40, 6.11, 1.33	0.63, 4.80, 1.70	0.27, 6.45, 1.25	0.59, 5.02, 1.62	0.64, 4.78, 1.74	0.61, 4.96, 1.66	0.52, 5.38, 1.55	0.48, 6.00, 1.49	0.59, 5.04, 1.63
Vegetated Soil Only	0.03, 19.95, 1.02	-0.11, 21.16, 0.97	-1.41, 30.05, 0.70	-0.07, 20.99, 0.99	0.00, 20.29, 1.01	-0.01, 20.41, 1.01	0.06, 19.62, 1.05	-0.01, 20.70, 0.99	0.04, 19.91, 1.04
Bare Soil + Raw Vegetated Soil	-0.01, 15.11, 1.02	0.04, 14.52, 1.09	-1.05, 21.02, 0.76	0.03, 14.90, 1.11	0.10, 14.50, 1.10	-0.03, 14.95, 1.05	0.03, 14.75, 1.05	0.08, 15.10, 1.04	0.00, 14.84, 1.04
ReflectGAN-Reconstructed Soil (Proposed)	0.32, 7.25, 1.50	**0.55, 5.35, 2.26**	0.44, 6.07, 2.11	0.55, 5.66, 2.00	0.56, 5.53, 2.09	0.54, 5.88, 1.91	0.49, 6.13, 1.81	0.36, 7.66, 1.46	0.53, 5.73, 1.99
Bare Soil + ReflectGAN-Reconstructed Soil	0.04, 9.29, 1.02	0.62, 5.56, 1.72	0.47, 6.57, 1.46	0.53, 6.29, 1.53	0.59, 5.87, 1.65	0.61, 5.85, 1.62	0.20, 9.20, 1.25	0.30, 9.31, 1.90	0.44, 8.26, 1.34

Overall, these findings demonstrate that ReflectGAN effectively mitigates vegetation-induced spectral contamination and enables accurate SOC mapping across heterogeneous landscapes containing both bare and vegetated surfaces, where traditional approaches typically struggle.

D. Limitations and Future Work

While this study highlights the effectiveness of ReflectGAN in improving SOC estimation through vegetation correction of satellite-derived reflectance, several important limitations should be recognized.

First, ReflectGAN relies on a paired dataset of vegetated and corresponding bare-soil reflectance, which may not always be feasible. In this study, barley-vegetated samples were statistically matched with their nearest bare-soil counterparts based on spatial proximity and spectral consistency. While this pairing strategy improves data reliability in controlled agricultural settings, it may limit scalability in heterogeneous landscapes where suitable bare references are sparse. To mitigate this limitation, future research could investigate unpaired or semisupervised GAN frameworks, such as domain adaptation GANs or cycle-consistent GAN architectures, that relax the requirement for strictly paired samples. Such extensions would enable the transfer of vegetation-to-soil correction knowledge to regions where temporally or spatially matched bare-soil observations are unavailable. Additionally, extending the approach to other types of vegetation would further test its robustness across diverse land cover conditions.

Second, the current model was trained and evaluated within a limited geographic scope and soil diversity. Although generalization was assessed using S2 imagery, broader validation across diverse agroecological zones is necessary to confirm its robustness. Evaluating the model across contrasting soil types, climatic regimes, and vegetation structures would provide stronger evidence of its transferability. At the same time, several characteristics of ReflectGAN support its potential generalization beyond the European context. The model was evaluated across two satellite sensors (L8 and S2) with differing spectral response functions and spatial resolutions, indicating robustness to sensor-specific variability. Moreover, ReflectGAN learns a vegetation-to-soil reflectance transformation rather than region-specific SOC relationships, and vegetation-induced spectral masking is governed by broadly consistent physical principles across landscapes. Consequently, the correction process is expected to be more transferable than direct SOC regression models, although broader multiregional validation remains an important direction for future research. Incorporating additional environmental variables such as soil moisture, texture, and seasonal dynamics could further improve the estimation accuracy and operational flexibility of the model.

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Third, while ReflectGAN improves the correlation between corrected reflectance and SOC, the reconstructed spectra may not always perfectly represent physically realistic soil characteristics. Integrating prior knowledge or physics-based constraints, such as hybrid physics-aware neural networks, could lead to more interpretable and scientifically consistent outputs. Moreover, future extensions could explore explainable DL frameworks and uncertainty-aware correction mechanisms to further enhance model transparency for operational soil monitoring.

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Finally, this study treated reflectance correction as a standalone preprocessing step for SOC estimation. Future research could explore end-to-end DL architectures that jointly optimize reflectance correction and SOC estimation within a unified framework, thereby reducing potential error propagation between stages. Additionally, incorporating temporal satellite observations and vegetation phenology information may further improve the robustness and reliability of SOC estimation across dynamic agricultural landscapes.

VII. Conclusion

This study introduced ReflectGAN, a novel paired-GAN framework specifically designed to reconstruct bare-soil reflectance from vegetation-contaminated satellite observations. By explicitly learning the transformation between vegetated and bare-soil reflectance, ReflectGAN effectively mitigates vegetation-induced distortion, providing cleaner inputs for SOC estimation. Extensive experiments on both L8 and S2 datasets demonstrated that ReflectGAN consistently outperformed traditional vegetation correction methods and existing GAN-based models, achieving significantly improved SOC estimation accuracy. Beyond its methodological contributions, ReflectGAN offers a scalable and computationally efficient solution for enhancing soil property mapping in vegetation-rich landscapes, paving the way for more reliable remote sensing applications in agriculture, land management, and environmental monitoring.

Data and Code Availability

The source code for the ReflectGAN model and the paired dataset (L8) used in this study are publicly available at: https://github.com/DristiDatta/ReflectGAN

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